Engineering Ethics Assignment Weeks 2-4

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Assignment Week 2 (300-500 words in total. Each topic 100-133 words on average.)

1. Are the nuclear attacks on Hiroshima and Nagasaki justifiable?

2. What do you find is the most major difference between the culture in your country and that in Japan?

3. How do you think your country would react if attacked by a nuclear bomb?

1)

The nuclear attacks on Hiroshima and Nagasaki violate human rights and rules of war on non-military personnels. The 10 rules of war, or Geneva conventions on international humanitarian law forbid attacking civil/non-military personnels. This is stated in Rule 3, "Prohibit targeting civilians. Doing so is a war crime." and Rule 1 "Protect those who are not fighting, such as civilians, medical personnel or aid workers." However, nuclear attacks destroy whole cities, communities, and harm so many innocent lives. 42 out of 45 hospitals were destroyed and several nurses were killed in Hiroshima. This damage on medical personnels and institutions violates Rule 6: " specify that medical workers, medical vehicles and hospitals dedicated to humanitarian work can not be attacked." People also got sick from radiation from the nuclear attack.

Last year, I listened to a first-person Nagasaki atomic bombing victim's testimony from Mrs. Matsumoto in my class. She was only 3 years old when the nuclear attack happened. What she went through was horrifying and traumatic. Not only she had lost her aunt, but also she was scared and traumatized for the rest of her life. This testimony made me realise how horrific nuclear attack is, and the importance of peace for the globe.

Nuclear bombing not only violates human rights, but also destroyed lives both physically and mentally. Therefore, nuclear attacks are not justifiable. The only method of war that is allowed is attack/defend against only military personnels.

Sources:

1. 10 things the rules of war do | International Committee of the Red Cross (icrc.org) , Internation Committee of the Red Cross
2. Hiroshima and Nagasaki bombings - ICAN (icanw.org)
3. Attending Matsumoto-san's testimony on Zoom for General Introduction to Global Issues class Week 4 (November/December 2023) and my assignment on the testimony

2) Japanese culture, with Ainu people and Okinawa people as major exceptions, is a relatively homogenous culture with few local dialects and few local languages. Most Japanese people is also a single ethnicity. 99.2% of people in Japan have Japanese as their first language. There are still a few lesser-spoken languages across the islands, including: Amami, Kyukyu, Kikai and Miyako. The Ainu language (spoken in Hokkaido) is critically endangered as a language, and only 15 people identified as speaking in the late 1990s.

Whereas the culture of Indonesia is diverse with 750 ethnic groups and over 700 local languages. The Austronesian are the biggest: the Javanese (Java) represent 40% of the total population; the Sundanese (West Java), the Madurese (Java and Bali), the Minangkabau (Sumatra) and Malays (Kalimantan) are also large groups. Another family, Papuans, number about a million. The Chinese arrived in Indonesia at the beginning of the Christian era and are a significant minority in the country. Major local languages are Javanese (32%), Sundanese (16%), Madurese (5%), Minangkabau (2%), Buginese (1.9%) , Palembang Malay (1.4%), and Banjarese (1.3%).

3) According to the Indonesian constitution and law, the country is against nuclear used for war purposes and intends to make Southeast Asia a nuclear weapon-free zone)-even gone so far as to strive not to make any nuclear military weapons even for defence to prevent any nuclear attacks. Indonesia is a also a neutral and traditionally non-aligned country.

Thus, as an anti-nuclear weapon and peaceful country, if there were a nuclear attack on Indonesia, Indonesia would:

a) Nuclear bombs devastate and damage whole cities and communities. As a vast country rich in natural resources, the population would be devastated. Nuclear bomb attack might also destroy transmission lines and transportation, so contact with the outside world might be difficult.

b) As Indonesia is currently not in war with any country and is a non-aligned country, Indonesia might be forced to enter the war to defend itself.
c) Indonesia is famously anti-nuclear weapon, so it might speak out in the United Nations condemning and opposing the use of nuclear bomb
d) Have to rebuild itself with the help of humanitarian aid from supporting and friendly countries

Sources: (Some are in Indonesian)

RI 'Dikepung' Senjata Nuklir, Ini Kata Pemerintah Jokowi (cnbcindonesia.com)

Buku Putih Pertahanan Indonesia 2008 (pdf) page 27

PENJELASAN ATAS UNDANG-UNDANG RE (kemenkeu.go.id)

Prabowo Warns of World War III Risk, Reaffirms Indonesia's Non-Aligned Stance

Assignment Week 3

List what is superior and what is inferior in the prevention of and the response to nuclear accidents by the Japanese nuclear engineers. (300-500 words)

A. Superior:
    1. Comprehensive safety improvements: Following the Fukushima Daiichi accident, Japanese nuclear engineers have implemented significant safety upgrades across all nuclear power plants. These include additional emergency power sources, fire engines, and reinforced on-site communication systems.
    2. Enhanced regulatory framework: Japan established a new regulatory regime, including the Nuclear Regulation Authority (NRA), to improve oversight and independence in nuclear safety regulation34.

Improved risk assessment: There's an increased focus on probabilistic risk assessment (PRA) and the evaluation of external hazards such as earthquakes and tsunamis.

    3. Emphasis on drill simulations
    4. Advanced equipment

B. Inferior:
    1. Failure of the plant owner (Tokyo Electric Power Company) and the principal regulator (Nuclear and Industrial Safety Agency) to protect critical safety equipment at the plant from flooding in spite of mounting evidence that the plant's current design basis for tsunamis was inadequate:

Some 18 years before the 2011 disaster, new scientific knowledge had emerged about the likelihood of a large earthquake and resulting major tsunami of some 15.7 metres at the Fukushima Daiichi site. However, this had not yet led to any major action by either the plant operator, Tepco, or government regulators, notably the Nuclear & Industrial Safety Agency (NISA). Discussion was ongoing, but action minimal. The tsunami countermeasures could also have been reviewed in accordance with International Atomic Energy Agency (IAEA) guidelines which required taking into account high tsunami levels, but NISA continued to allow the Fukushima plant to operate without sufficient countermeasures such as moving the backup generators up the hill, sealing the lower part of the buildings, and having some back-up for seawater pumps, despite clear warnings. (Source : Fukushima Daiichi Accident - World Nuclear Association)

In 2009, there was also a warning to TEPCO about a possibility that a tsunami could happen to Fukushima. TEPCO was still regarding the matter when the natural disaster happened.

2. At first, the company that operates the plant, TEPCO, did not allow its workers to speak freely and publicly about what happened in the Fukushima earthquake and the nuclear reactor disaster. Only a year later did their stories began to emerge.

Assignment Week 4

What level of education on nuclear physics is needed for workers in nuclear facilities? (300-500 words)

Entry-level and support staff:

For many entry-level positions and support roles, a high school diploma or equivalent is the minimum requirement. However, these workers still need to undergo extensive on-the-job training to develop "nuclear awareness" - a basic understanding of nuclear safety principles and the unique work environment

Technicians and operators:

Nuclear plant operators and technicians typically need at least a high school diploma, though many employers prefer candidates with some college education or an associate's degree in a related field. These workers must complete rigorous training programs

lasting 18 to 24 months, which include both classroom instruction and hands-on experience with nuclear physics principles

Engineers and specialists:

For engineering and specialist roles, a bachelor's degree is typically the minimum requirement. Preferred fields of study include nuclear engineering, physics, chemistry, or other related engineering disciplines. These "nuclearised" professionals need to acquire in-depth knowledge of the nuclear environment to apply their competencies effectively. Nuclear engineers typically need at least a bachelor’s degree in nuclear engineering or a related field. They typically do not need a license to enter the occupation.

Advanced positions and research roles:

Senior positions, research roles, and specialized technical jobs often require advanced degrees. A master's degree in nuclear science and technology or a related field is common for these positions. Some roles, particularly in research and development or academia, may require a Ph.D. in nuclear physics or engineering